Examiner only

7. (a) In the table below, statements are given which describe different particles or interactions.
Complete the table by naming the particle or interaction described. [4]

Description of particle or type of interaction	Name of particle or interaction
The quark combination of this particle is u u d.	....
The electron and electron neutrino are examples of this group of particles.	....
Antibaryons are a combination of three of these.	....
Neutrino involvement and quark flavour changes are exclusive to this type of interaction.	....

(b) An antiparticle has a quark composition of $\bar{u}\,\bar{d}\,\bar{d}$. Determine its charge and identify the particle. Show your working clearly. [2]

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(c) (i) Consider the following hypothetical interaction.

$$p + n \longrightarrow p + \bar{p} + p + \nu_e$$

The reaction is **not possible** because it does not obey one or more of the conservation laws. By considering baryon number, lepton number and charge, show which law(s) are obeyed and which are not. [3]

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(ii) Jon suggests that replacing the electron neutrino (ν_e) with a pi-zero pion (π^0) would allow the reaction to occur. Explain whether or not he is correct. [1]

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